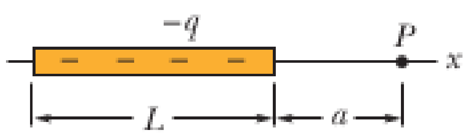
Department of **Mathematics and Natural Sciences**
PHY112 - Principles of Physics II
Assignment 2
Total marks: 15

Question 1

Figure shows a nonconducting rod of length L = 5.5 cm has a nonuniform charge density $\lambda = -2x$.

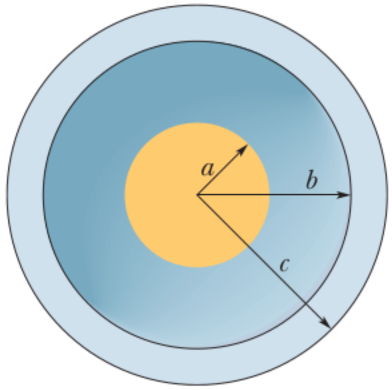


(a) Find the Potential at point P which is at a distance a = 10 cm from the rod. 4

(b) The electric potential at points in an xyz plane is given by V= (5.0 V/m^2) x^2y + (4.0 V/m^4) y^3z+ (6.0 V/m^3) yz^2. In unit-vector notation, what is the electric field at the point (3.0 m, -2.0 m, 9 m) and the magnitude of the field? 4

Question 2

The figure shows a solid sphere of radius a = 3.00 cm is concentric with a spherical conducting shell of inner radius b = 2.00a and outer radius c = 3.0a. The sphere has a net uniform charge $q_1 = 5.00$ pC, the shell has a net charge $q_2 = -4q_1$.



Using Gauss's law Find:

(a) What would be the electric field at r = a/2 and r = 1.7a and the change density in the inner shell and outer shell? 4

(b) Using Gauss law find out what would be the electric flux at r=3a and r = 3.5a. 3